Unit-I

1. Define Operating System and explain the various types of Operating Systems?
2. Explain Operating System Structures?
3. Explain the different functions of an operating system and discuss the various services provided by an operating system.
4. Write the differences between monolithic kernel and microkernel.
5. Explain briefly system calls with examples.
6. Explain different types of system calls with suitable example.
7. What are the functionalities of Operating Systems? Explain in detail
8. Explain difference between Multitasking and Multi Programming?
9. Differentiate between Batch Operating System and Time Sharing Operating System?
10. List out types of operating system. Explain operating system services.
11. Explain the structure of operating system.
12. What is spooling? Explain with example.

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Unit-II

1. Define Process? Explain process State diagram?
2. Explain about process schedulers?
3. Consider 3 processes P1, P2 and P3, which require 5, 7 and 4 time units and arrive at time 0, 1 and 3. Draw the Gant chart, process completion sequence and average waiting time for. (i) Round robin scheduling with CPU quantum of 2 time units. (ii) FCFS.
4. Explain CPU Scheduling Algorithms with examples?
5. Explain about Scheduling Criteria.
6. Assume arrival order is: P1, P2, P3, P4, P5 at time 0,1,2,3,4 respectively. They have estimated running times 10,6,2,4 and 8 minutes. Draw the Gantt chart for FCFS, RR and SJF(Preemptive). Time quantum is 2ms.
7. What is Process? Give the difference between a process and a program.
8. Explain the Process State transition diagram. Explain Process Control Block (PCB)?
9. Difference between process and thread.
10. What is scheduler? Explain queuing diagram representation of process scheduler with figure.
11. Five batch jobs A to E arrive at same time. They have estimated running times 20,12,6,8,4 minutes. Their priorities are 3,2,5,4,1 respectively with 5 being highest priority. For each of the following algorithm determine average process turnaround time. Round Robin (q=3), Priority Scheduling, FCFS, SJF.
12. Write the difference between user level thread and kernel level thread?
13. Evaluate FCFS, RR, Priority and SJF (Preemptive) CPU Scheduling algorithm for given Problem. Consider lowest value as higher priority.

Process	Arrival Time	Burst Time	Priority
P1	0	8	4
P2	1	4	3
P3	2	9	1
P4	3	5	2

14. Consider the following set of jobs. Determine the finish & turnaround time for the above jobs using 1. FCFS 2. SJF 3.Round Robin(quantum time 0.5)

Job No.	Arrival Time	Run Time
1	10	2
2	10	2
3	11	1
4	13	3
5	14	1

Also solve with priority scheduling algorithm having priority 8, 2, 6, 5, 4.

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Unit-III

1. What is deadlock? What are the criteria's for occurrence of a deadlock?
2. Solve the below example using Banker's Algorithm

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

3. What is critical section problem? Explain with example?
4. What is Semaphore? Explain producer consumer problem using semaphore?
5. Define process synchronization and explain Peterson solution algorithms?
6. What is Monitor? Explain with any example using monitor?
7. Explain the solution for Dining-Philosophers Problem
8. What are the methods for handling deadlock?
9. Write about deadlock and starvation?
10. Explain about Deadlock Avoidance?
11. Explain how recovery from deadlock?
12. Explain Dead lock detection (Banker's Algorithm) with Example?
13. Write about Deadlock Prevention Methods?
14. Discuss about the following
    a. Semaphore
    b. Monitor
15. What is inter process communication? Explain it's methods.
16. What is Mutex? Explain with any example using mutex?
17. Considering a system with five processes P0 through P4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Solve the following example with Banker's algorithm.

	Allocation	*Max*	*Available*
	A B C	*A B C*	*A B C*
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

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Unit-IV

1. Explain the types of fragmentation and ways to reduce them.
2. How operating system utilizes swapping technique in memory management? Explain in detail.
3. Consider the following page reference string: 8,1,3,5,3,1,4,8,5,7,2,1,2. The no. of page frames=3 calculate the page faults and the hit ratio FIFO, Optimal, LRU page replacement algorithm.
4. Explain segmentation with paging memory management technique.
5. List the page replacement algorithms and explain any one.
6. What is Memory Management Unit? Explain functionalities of memory management unit.
7. Explain memory relocation and protection in detail
8. Explain memory hierarchy in detail with diagram.
9. Write down the difference between Logical and Physical Address.
10. Explain Memory Management with Dynamic-Partitions.
11. Explain Memory Management with Fixed Partitions.
12. Write short note on Fragmentations.
13. Differentiate external fragmentation v/s internal fragmentation.
14. Explain Virtual Memory.
15. Explain Paging memory management technique.
16. Explain segmentation memory management technique.
17. Write short note on demand paging
18. Given page reference string: 1,2,3,2,1,5,2,1,6,2,5,6,3,1,3,6,1,2,4,3. Compare the number of page faults for LRU, FIFO and Optimal page replacement algorithm.
19. What is virtual memory? Discuss the benefits of virtual memory techniques.
20. Write short notes on
    a) Demand paging
    b) Thrashing
    c) Page replacement

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Unit-V

1. Explain file management.
2. Write a short note on file structure.
3. Write a short note on directory structure.
4. Explain all disk scheduling policies.
5. Explain the different Disk scheduling algorithms with their comparisons.
6. Suppose that a disk drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at cylinder 143. The queue of pending requests in FIFO order 86,1470,913,1774,948,1509, 1022, 1750, 130 starting from current head position. What is
   the total distance that disk arm moves to satisfy all the pending request for FCFS and SSTF disk scheduling algorithm.
7. Explain the following disk scheduling algorithm with proper diagram
   a) FCFS
   b) SSTF
   c) SCAN
   d) LOOK
   e) C-SCAN.
8. Explain about different directory structure available.
9. What are the Access methods available?
10. Explain the concept of file with Example.
11. Explain about file access methods with Example.
12. Write about different types of operation performed on file.
13. Explain in detail file allocation methods.
14. Explain the file systems in operating system.
15. Write notes about disk management.

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